

AGENTIC RL · DEEP RESEARCH

# LiteResearcher

*A Scalable Agentic RL Training Framework  
for Deep Research Agents*

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[arXiv:2604.17931](https://arxiv.org/abs/2604.17931) · [github.com/simplex-ai-inc/LiteResearcher](https://github.com/simplex-ai-inc/LiteResearcher)

Zhejiang University · Simplex AI · HKPU

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# Authors, links & one-line summary

## AUTHORS

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## RESOURCES

**Paper** · **arXiv:2604.17931**

Code · [github.com/simplex-ai-inc/LiteResearcher](https://github.com/simplex-ai-inc/LiteResearcher)

Weights · [huggingface.co/simplex-ai-inc/LiteResearcher-4B](https://huggingface.co/simplex-ai-inc/LiteResearcher-4B)

Venue · *Preprint, under review*

## TL;DR

*A virtual world that mirrors live web search but isolates its noise lets a tiny 4B model match or beat 8× larger open-source and commercial agents on deep-research benchmarks.*

# What we will cover today

01

## Problem & prior approaches

Why scaling Agentic RL for deep research is hard

02

## Method overview

The 3-pillar LiteResearcher pipeline

03

## Pillar 1 · Data × Corpus

Co-construct synthetic data and local corpus

04

## Pillar 2 · Local tool environment

32M-page virtual web at <0.2 s latency

05

## Pillar 3 · Curriculum RL

On-policy GRPO + difficulty filtering

06

## Main results

GAIA, Xbench, BrowseComp, HLE, ...

07

## Ablations

Cost, data, on/off-policy, two-stage, RL vs SFT

08

## Behavior analysis · appendix

Four mechanisms behind the Sim2Real gain

# Why scaling Agentic RL for Deep Research is hard

*“ RL has delivered sustained gains for closed-world reasoners; Agentic RL for deep research has not. Why? ”*

## A · DATA BOTTLENECK

### Hand-crafted synthetic data

Over-engineered task templates that do not elicit the diverse atomic search skills used in the real world  
— cross-verification, enumeration, statistics, ...

## B · ENVIRONMENT BOTTLENECK

### Real-web RL dependency

Live-internet rollouts inject high variance and prohibitive cost into RL training, capping the number of stable updates and the scale of training data.

# Three families, three trade-offs

## WEBTHINKER · KIMI-RESEARCHER

### Online RL

*Live internet · high realism*

#### STRENGTHS

- + Realistic interactions
- + Unbounded knowledge

#### LIMITATIONS

- Non-deterministic reward
- Cost grows with steps
- Variance bottlenecks scale

## SEARCH-R1 · SEARCH-O1

### Local retrieval

*Wikipedia-scale corpus*

#### STRENGTHS

- + Cheap, fast, deterministic

#### LIMITATIONS

- Narrow, homogeneous
- Misses real-web dynamics
- Caps the search-skill ceiling

## ZEROSEARCH

### Simulated search

*LLM-mimicked search engine*

#### STRENGTHS

- + Fully controllable

#### LIMITATIONS

- No page-level fidelity
- Hallucinated documents
- Limited transfer to real web

# Mirror the web, isolate the noise

## OUR ARGUMENT

Scaling Agentic RL for Deep Research requires a twin architecture:

**Mirroring** Local environment behaves like the live web — same page-level fidelity, same retrieval distributions, same browse dynamics.

**Isolation** Training is decoupled from the real internet — no rate limits, no API cost, no non-deterministic reward, no rollout variance.

## OUTCOME

### LiteResearcher-4B

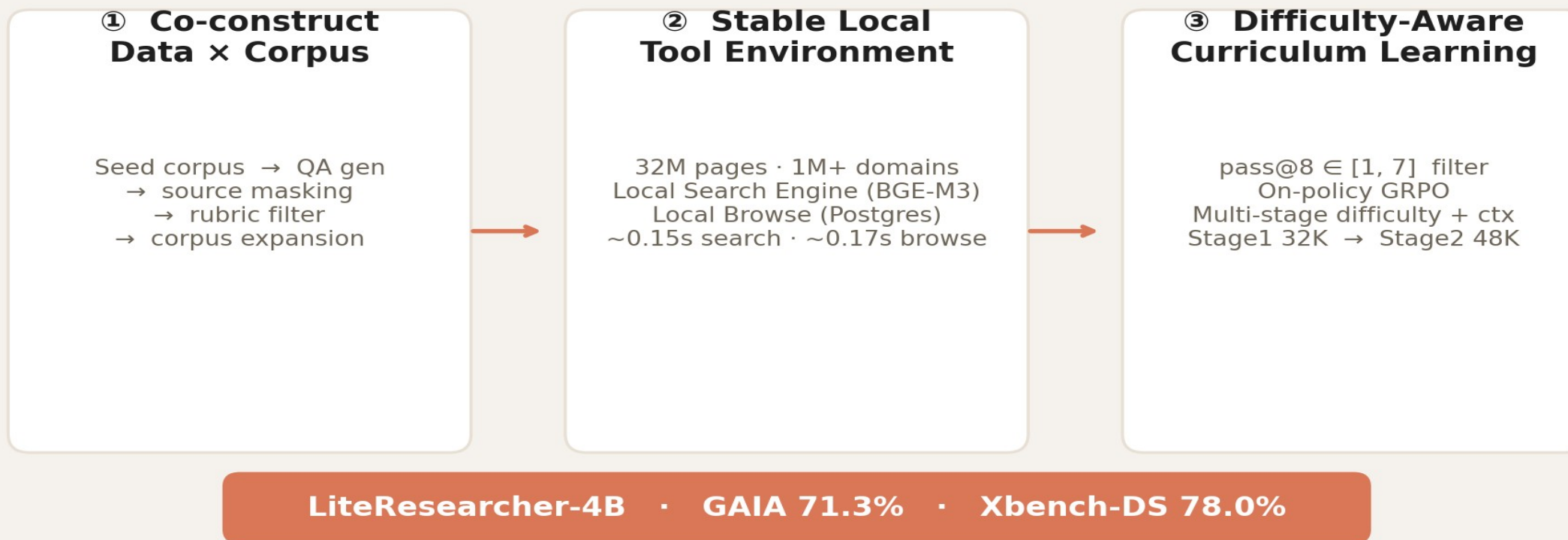
*A 4B model that scales like a 30B+ one*

**71.3%** GAIA

**78.0%** Xbench-DS

**83.1%** Frames

# Three pillars of the LiteResearcher pipeline



# Co-construct training data and local corpus

## WHY CO-CONSTRUCTION

Tasks and corpus evolve together so that every synthetic query is answerable by browsing the local environment — no shortcut, no leak.

### STEP 01

#### Seed Corpus

Wikipedia + BBC News  
→ structured QA

### STEP 02

#### Source Masking

Delete the source page  
so the answer is non-trivial

### STEP 03

#### Rubric Filtering

7-criteria LLM filter  
(independence,  
verifiability, ...)

### STEP 04

#### Corpus Expansion

Fetch related web pages  
→ enrich the local pool

# Five atomic search capabilities

*The taxonomy of skills our data must elicit*

**01**

## Direct Information

Single fact, single retrieval.  
e.g. "When was X founded?"

**02**

## Aggregation

Find target by multiple  
identifying attributes.

**03**

## Enumeration

List & count entities that  
satisfy a constraint.

**04**

## Cross-verification

Validate claims across  
multiple independent sources.

**05**

## Statistics

Aggregate numerical metrics  
from multiple pages.

# Stable local tool environment

## THE VIRTUAL WEB

**32 M**

pages

**1 M+**

domains

**73.2 M**

tool calls  
over full RL run

### INDEX · RETRIEVE

## Local Search Engine

BGE-M3 hybrid dense + sparse  
Milvus index · DiskANN on disk

~0.15 s / query (≈ 10× faster than online)

### STORE · SERVE

## Local Browse Tool

Full Markdown content in PostgreSQL  
Keyed by URL · tuned for 1,000 concurrent conn.

~0.17 s / page (≈ 46× faster than Jina Reader)

# Difficulty-aware curriculum learning

## THE PROBLEM · TRAINING SATURATION

When too easy → no gradient. When too hard → no signal.  
Single-stage RL plateaus at the model's current difficulty band.

## THE FIX · TWO LEVERS

### Difficulty filter

Before each stage:  $K=8$  rollouts per query, keep only those with correct-count  $c \in [1, 7]$ . Discard trivial and impossible.

### Multi-stage curriculum

Each stage → increase task difficulty + context length  
(Stage 1 = 32K → Stage 2 = 48K, with harder data mix).

## ON-POLICY GRPO

### The training objective

$$J(\theta) = E \left[ \frac{1}{K} \sum \min \left( r_i \cdot A_i, \text{clip}(r_i, 1-\epsilon_{\text{lo}}, 1+\epsilon_{\text{hi}}) \cdot A_i \right) \right]$$

$$r_i(\theta) = \pi_{\theta}(o_i | q) / \pi_{\text{old}}(o_i | q)$$

- No KL penalty
- No entropy regularizer
- Strictly on-policy: each rollout used once

# 06

## EXPERIMENTS

# Main results & ablations

*8 benchmarks · ablations on data, environment,  
on/off-policy, multi-stage, SFT-only*

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# Training recipe at a glance

## RL TRAINING CONFIGURATION

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| Base model     | <b>Qwen3-4B-Thinking-2507</b>                                            |
| Cold-start SFT | <b>trajectories distilled from Tongyi DeepResearch</b>                   |
| Batch size     | <b>global 128 · 8 rollouts per query · on-policy</b>                     |
| Learning rate  | <b><math>1 \times 10^{-6}</math> (constant)</b>                          |
| Loss           | <b>GRPO · no KL · no entropy regularizer</b>                             |
| Context length | <b>Stage 1 = 32 K → Stage 2 = 48 K</b>                                   |
| Eval tools     | <b>Serper for search · Jina Reader for browse (online APIs)</b>          |
| Benchmarks     | <b>GAIA · Xbench-DS · BrowseComp · HLE · Frames · WebWalker · Seal-0</b> |

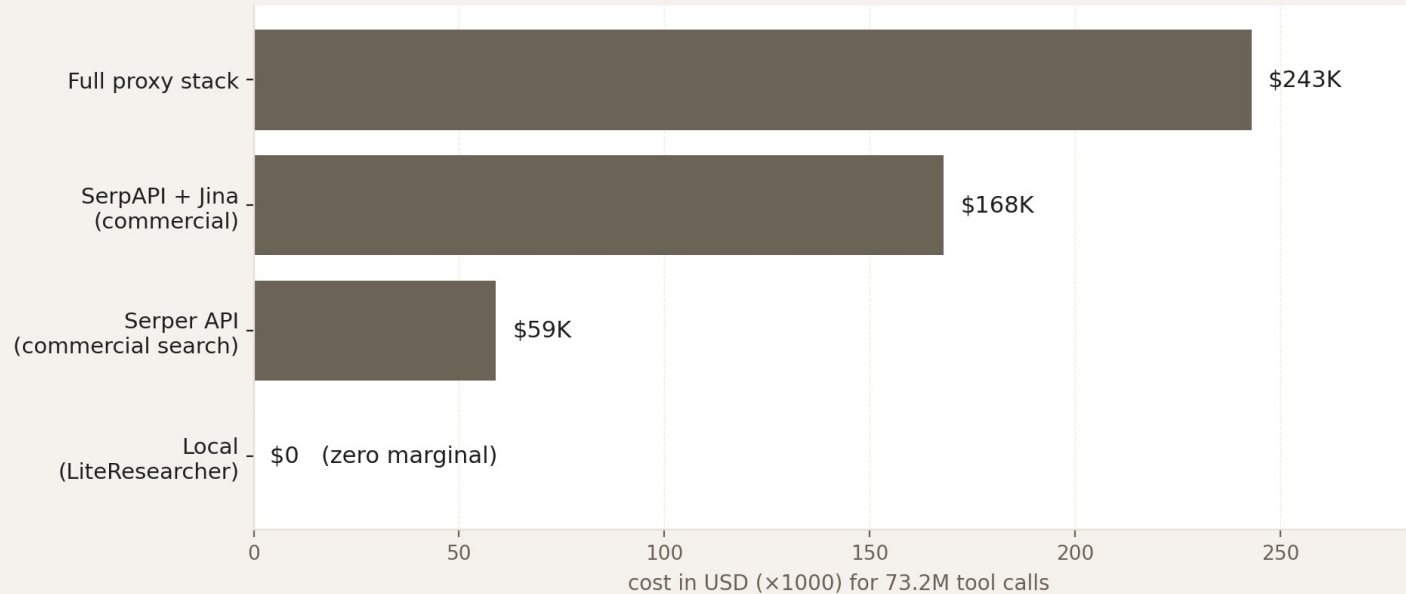
# 8 benchmarks · 4B model · open-source SOTA

| Model                        | Size      | GAIA        | Xbench-DS   | Frames      | BrowseComp  |
|------------------------------|-----------|-------------|-------------|-------------|-------------|
| Claude-4.5-Sonnet            | —         | 71.2        | 66.0        | 80.7        | —           |
| OpenAI GPT-5-high            | —         | —           | 77.8        | —           | —           |
| GLM-4.6                      | —         | 71.9        | 70.0        | —           | —           |
| DeepSeek-V3.2                | —         | —           | —           | 80.2        | —           |
| Tongyi DeepResearch          | 30B       | 70.9        | 75.0        | —           | —           |
| WebSailor                    | 30B       | 53.2        | 53.3        | —           | —           |
| AgentCPM-Explore             | 4B        | 63.9        | 70.0        | —           | —           |
| <b>LiteResearcher (ours)</b> | <b>4B</b> | <b>71.3</b> | <b>78.0</b> | <b>83.1</b> | <b>32.5</b> |

All scores in %. LiteResearcher-4B matches or exceeds models 8x its size and most commercial systems.

**4 B beats 30 B · open-source SOTA on GAIA · Xbench · Frames · zero marginal cost during training**

# Local vs. live internet — cost & latency



## TOTAL TOOL CALLS

**73.2 M**

45.8 M search + 27.4 M browse  
over the full RL run

## SPEED-UP

**10× faster search**

**46× faster browse**

*Throughput is what makes  
scalable on-policy RL feasible.*

# Effect of our synthetic data

## DOES SCALING INFORMATION SOURCES PAY OFF?

| Training data                    | GAIA        | Xbench-DS   |
|----------------------------------|-------------|-------------|
| Multi-hop data only              | 58.7        | 66.3        |
| <b>Multi-hop + our synthetic</b> | <b>66.8</b> | <b>71.0</b> |

## INTERPRETATION

Adding our synthetic data lifts both benchmarks by +8.1 / +4.7 points.  
A scaled-and-filtered information source covers the long-tail of search patterns that hand-crafted multi-hop data misses.

# On-policy vs. off-policy GRPO

## SETUP

Off-policy variant: each rollout batch (256) is split into 4 mini-batches and consumed in 4 successive updates.

On-policy: each rollout batch is consumed in a single update and discarded.

## RESULT

Off-policy gains reward earlier but eventually declines.

On-policy keeps improving — higher final GAIA validation.

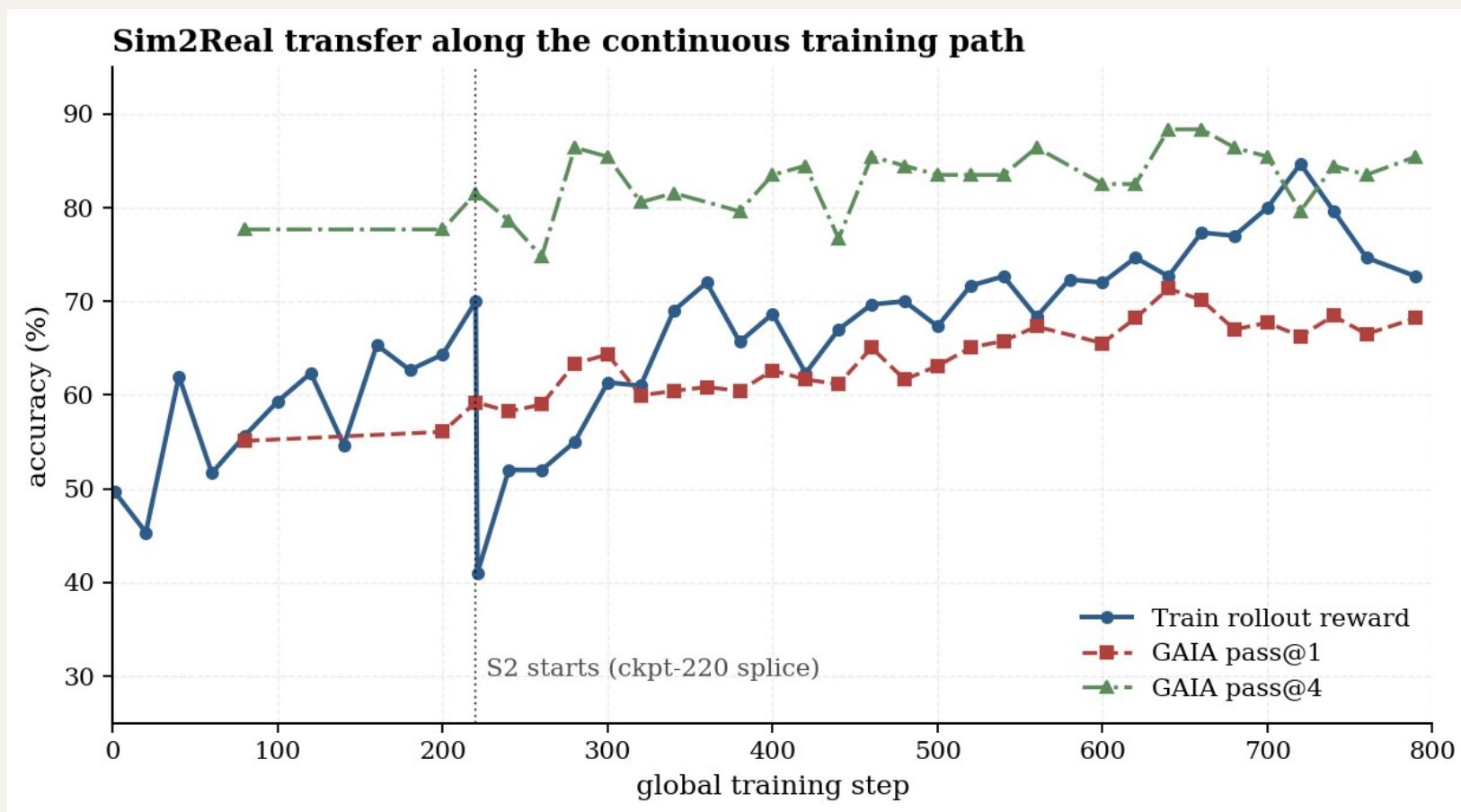
### GAIA VALIDATION ACCURACY

|            |        |
|------------|--------|
| Off-policy | 66.8 % |
|------------|--------|

|           |        |
|-----------|--------|
| On-policy | 68.9 % |
|-----------|--------|

*Multiple updates per batch → policy lag → trajectory mismatch → ceiling drops.*

# Effectiveness of multi-stage training



GAIA · EMA-SMOOTHED

Stage 1 plateau

**64.7 %***at ~step 220*

Stage 2 end

**68.3 %***+3.6 pts*

*Stage 2 lifts the plateau by  
raising difficulty + context length.*

# Contribution of RL over SFT

SFT TEACHER · TONGYI DEEPRESEARCH (30B)

TEACHER (30B)

**70.9 %**

*where the teacher caps*

SFT ONLY (4B)

**55.6 %**

*−15.3 pts vs teacher*

+ RL (4B, OURS)

**71.3 %**

*+15.7 pts over SFT*

*The primary performance driver is the RL framework — not teacher distillation.*

# 08

## Why does the policy improve?

A behavior-level autopsy of 41 checkpoints across  
S1 (steps 0-220) → S2 (steps 0-570), spliced at  
step 220.

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*30 metrics · 300 traj/step · GAIA every 20 steps*

# What changed about the model's behavior?

## MOTIVATION

The aggregate reward curve goes up — but what does the policy

*actually do differently at step 790 vs step 0?*

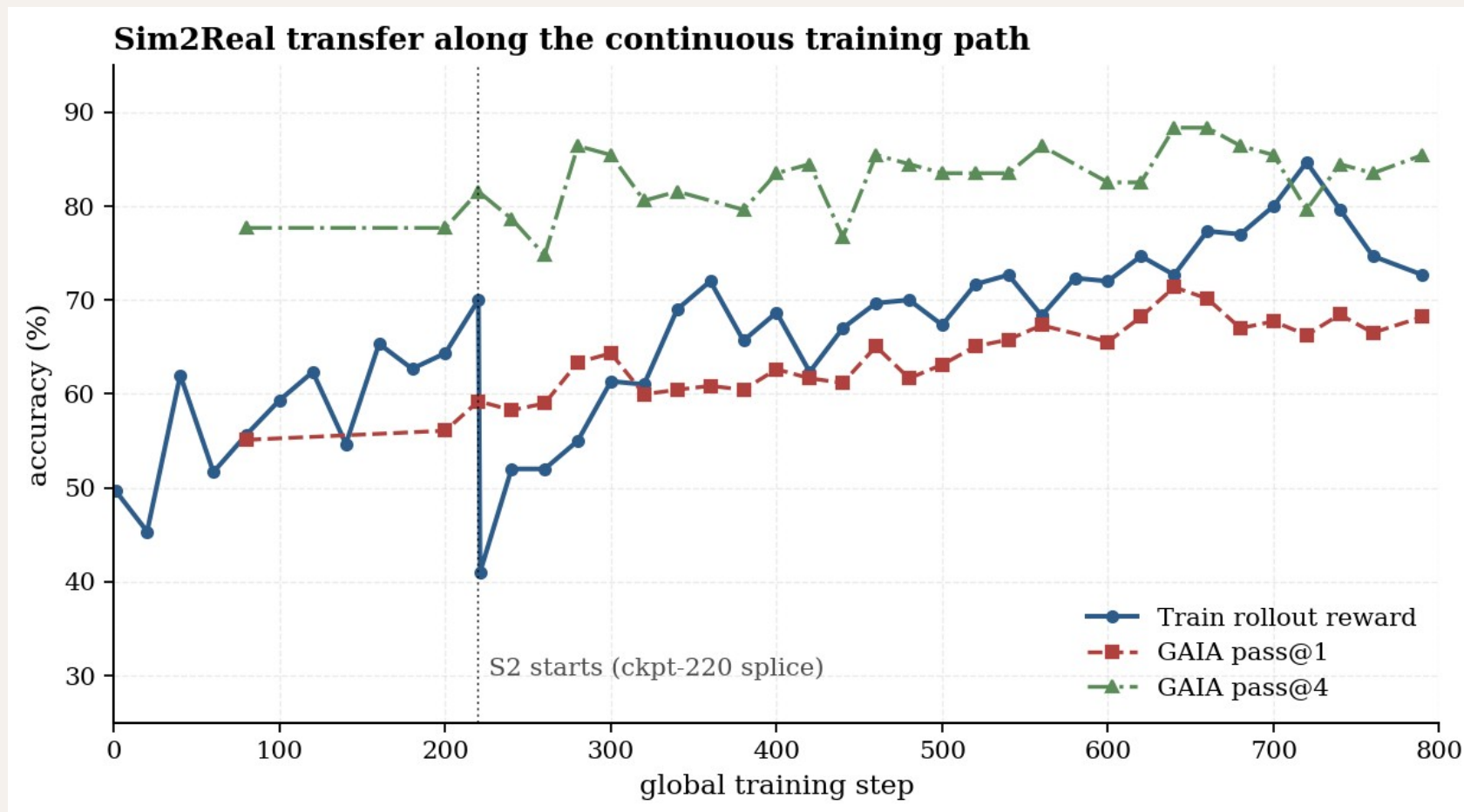
We log 30 per-trajectory behavior signals at every checkpoint, then look for monotone shifts that correlate with the GAIA accuracy gain.

***Goal: isolate causal patterns from incidental noise.***

## ANALYSIS SETUP

|                             |                                     |
|-----------------------------|-------------------------------------|
| Checkpoints sampled         | <b>41</b>                           |
| Train rollouts per ckpt     | <b>300 trajectories</b>             |
| GAIA-eval rollouts per ckpt | <b>412 tasks × pass@1, pass@4</b>   |
| Behavior metrics extracted  | <b>30 per trajectory</b>            |
| Splice point                | <b>step 220 — S1 end → S2 start</b> |
| Total trajectories analyzed | <b>~ 24,000</b>                     |

# What we measured at the outcome level



## OBSERVATION

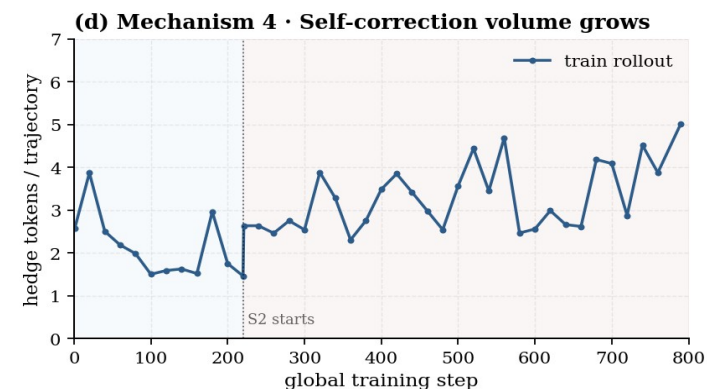
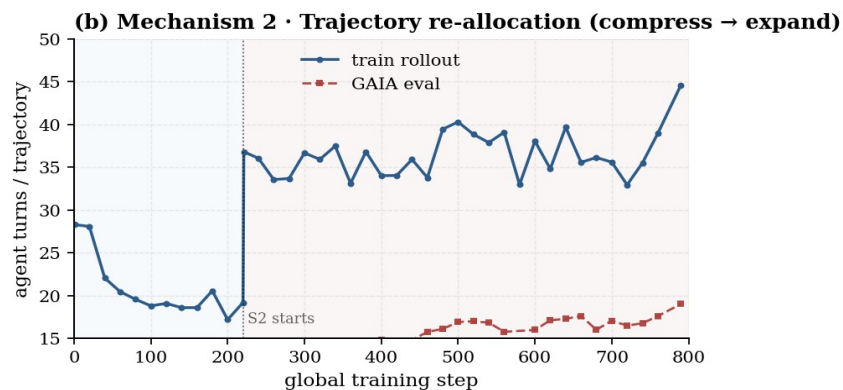
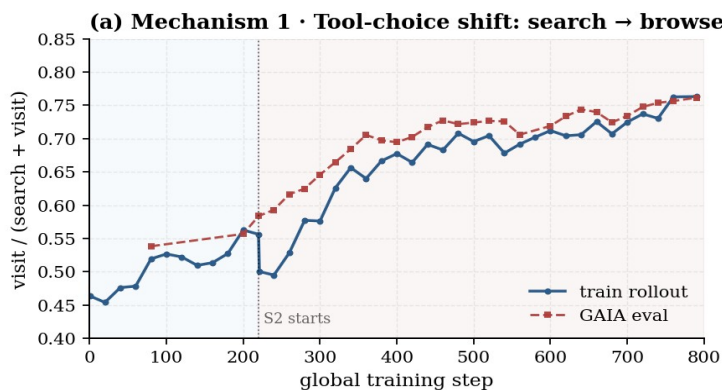
**Train reward and GAIA evaluation co-improve through both stages.**

S1 saturates near step 220 — S2 lifts the plateau without regression.

→ *Sim2Real transfer is sustained.*

# Four mechanisms behind the gain

Four behavioral mechanisms underlying the Sim2Real gain (S1 0-220, S2 0-570 spliced at step 220)



Blue = train rollout (300 traj/step). Red = GAIA eval (412 tasks). Panel (d) is train-only because GAIA logs strip <think> blocks.

## THE FOUR LEVERS

### M1 Tool-choice shift

*search → browse*

**0.46 → 0.76**

### M2 Trajectory re-alloc

*compress → expand*

**28 → 19 → 45**

### M3 Per-think depth

*richer reasoning blocks*

**387 → 626 chars**

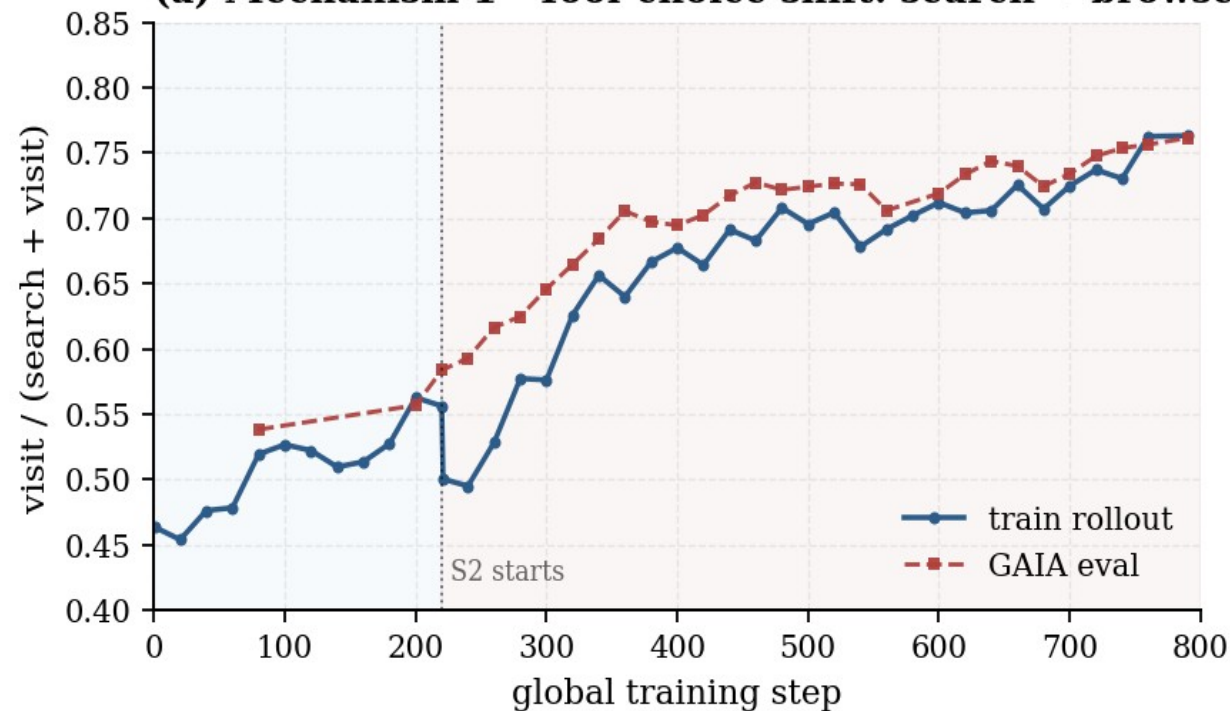
### M4 Self-correction

*more hedge volume*

**1.5 → 5.0 tokens**

# From keyword search to deep browse

(a) Mechanism 1 · Tool-choice shift: search → browse



browse / (search + browse)

step 0 **0.46**

step 220 **0.56**

step 460 **0.72**

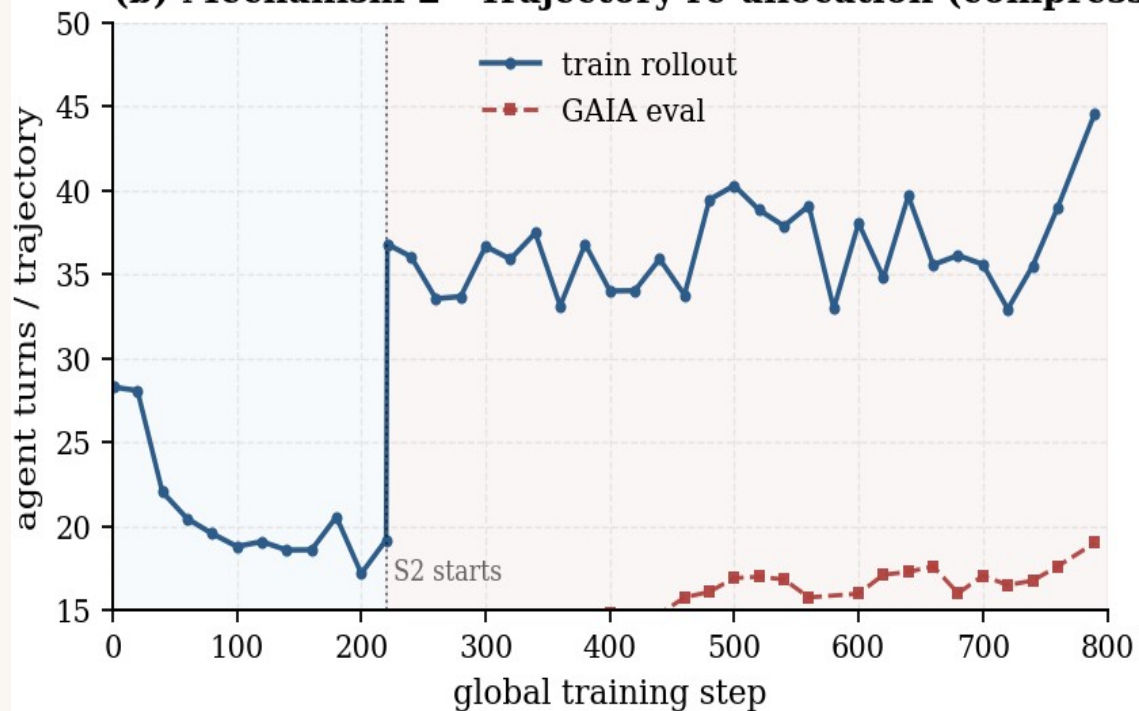
step 790 **0.76**

## INTERPRETATION

Policy stops treating search as the answer and starts using it as a pointer toward evidence — which it then opens and reads.

# Compress, then expand into harder problems

(b) Mechanism 2 · Trajectory re-allocation (compress → expand)



agent turns / trajectory

**28 → 19 → 45**

*S1 start · S1 end · S2 end*

## Phase A · S1 (0→220)

Policy learns to compress —  
fewer wasted turns per task.

## Phase B · S2 (220→790)

Same policy meets harder tasks  
and an enlarged 48k context — it  
re-invests the budget for depth.

# Each reasoning block carries more content

chars / <think> block

**387 → 626**

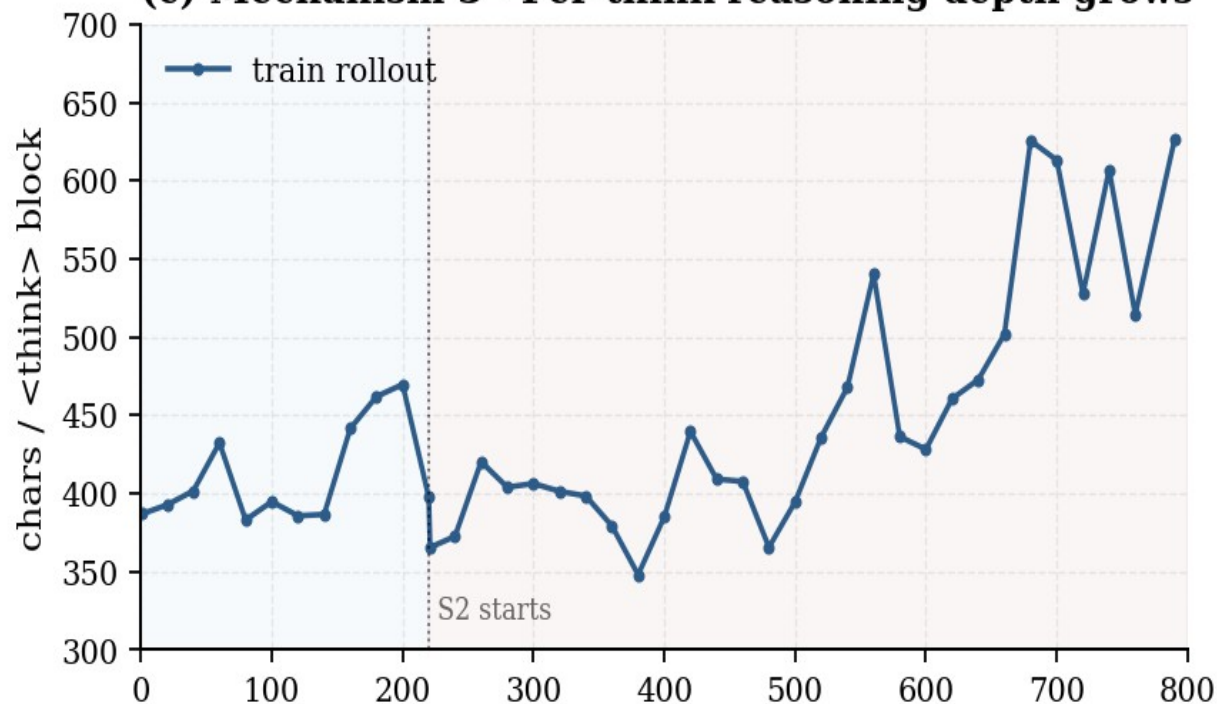
≈ +62 % per reasoning block

## WHY DEPTH, NOT FREQUENCY?

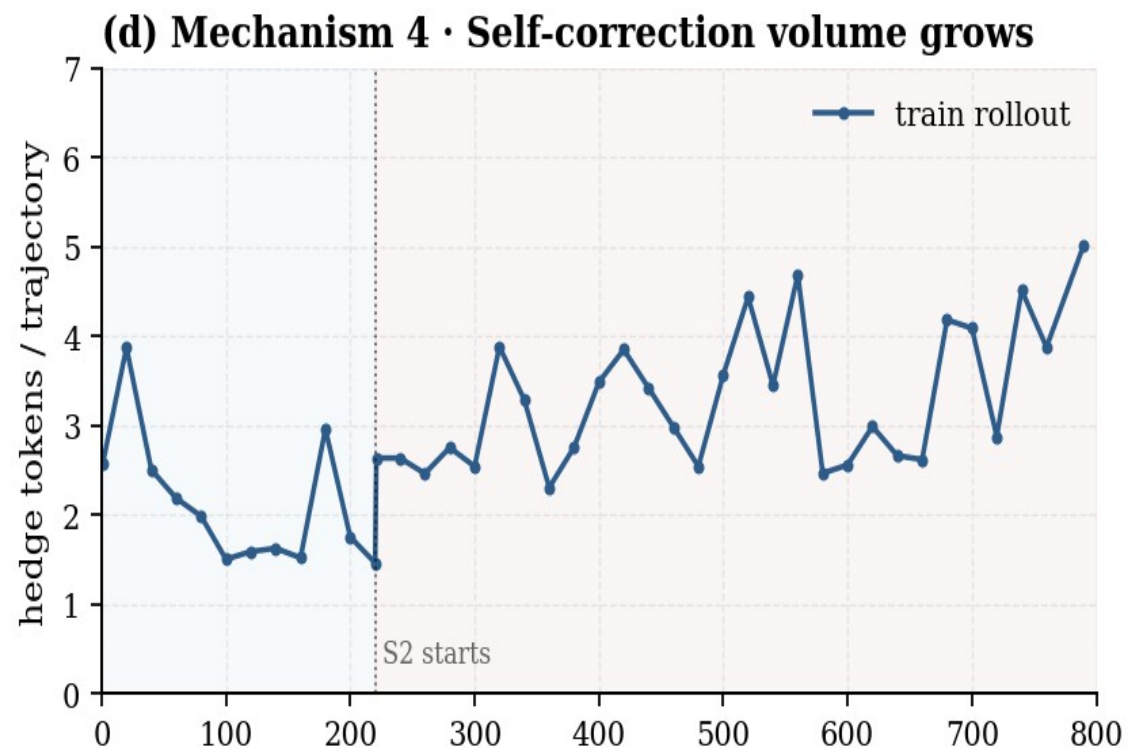
Number of <think> blocks per turn stays flat — what grows is the content inside each block.

*The policy reasons more carefully, not more often.*

(c) Mechanism 3 · Per-think reasoning depth grows



# More hedging and verification moves



hedge tokens / trajectory

**1.5 → 5.0**

*"wait / verify / actually / let me check"*

## HONEST FRAMING

Hedges per token barely changes —  
the trajectory simply gets longer  
and contains more verification.

*Self-correction is a volume effect,  
driven by the expanded budget.*

# Candidates we ruled out — for transparency

## RULED OUT BY THE DATA

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| <b>Answer-length growth</b>      | Confounded with task difficulty in S2      |
| <b>Tool calls per turn</b>       | Stable across both stages — no shift       |
| <b>Think frequency per turn</b>  | Flat — the depth, not the count, grows     |
| <b>Hedge density (per token)</b> | Almost unchanged — only volume grows       |
| <b>Vocabulary diversity</b>      | No monotone trend across checkpoints       |
| <b>Error/retry rate</b>          | Drops in S1, rebounds in S2 — non-monotone |

## CAUSAL CHAIN

**M1** **Tool-choice**  
*search → browse*

**M2** **Re-allocation**  
*expand budget*

**.** **More evidence**  
*per trajectory*

**M3+** **Deeper reason &**  
**M4** *self-correction*

→ **GAIA +9 pts**  
*(60 → 69)*

# Three things to remember

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01

## **Local environments are enough.**

A 32 M-page Wikipedia mirror + a lightweight browse tool replaces \$59 K-\$243 K of commercial APIs, gives 10×/46× speed-ups, and unlocks the trajectory volume needed for on-policy RL at scale.

02

## **Simple-but-scalable beats clever-but-fragile.**

Atomic-capability synthesis, on-policy GRPO, and a 2-stage curriculum — each individually familiar — combine to lift a 4 B model past 30 B teachers and most commercial systems.

03

## **The gain has a behavioral signature.**

Two structural mechanisms (tool-choice shift, trajectory re-allocation) act first; depth and self-correction emerge as their downstream effect. Sim2Real is sustained because the behavior, not just the score, transfers.

# Thank you

*Questions & discussion welcome.*

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## PAPER

arXiv : 2604.17931

## CODE

[github.com/wanli/LiteResearcher](https://github.com/wanli/LiteResearcher)

## MODEL

[hf.co/litererresearcher-4b](https://huggingface.co/litererresearcher-4b)

*Wanli Lee · with thanks to all co-authors and reviewers*